A General Approach to Terminal Allenols

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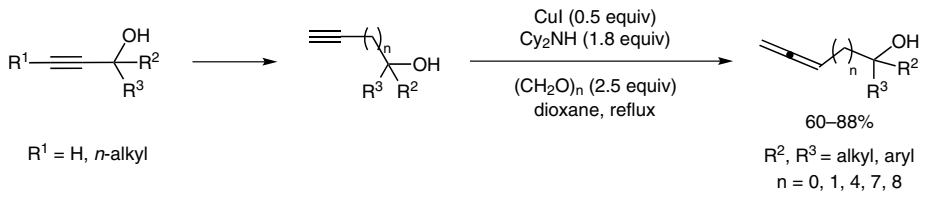
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Abstract: We have demonstrated a very general method for the preparation of essentially any terminal 2,3-allenol from the corresponding alkynols, which may be easily available from propargylic alcohols by alkylation, or from terminal alkynes by deprotonation and 1,2-addition with aldehydes or ketones, and subsequent base-catalyzed triple-bond migration.

Key words: copper(I) iodide, dicyclohexylamine, allenols, long carbon chain, chirality



Scheme 1 CuI-mediated synthesis of terminal allenols with dicyclohexylamine

Introduction

During the last 10–15 years, allenes have been shown to be versatile intermediates in organic synthesis.[1,2] In addition to this, many natural products and pharmaceuticals with an allene moiety have been identified.[3] Thus, synthetic methods for allenes from readily available organic compounds[4] are of high interest for chemists who use allenes as the starting materials for their target syntheses. Crabbé and co-workers reported the first CuBr-mediated reaction to form terminal allenes from 1-alkynes and formaldehyde in the presence of diisopropylamine.[5a–d] Based on this, recently we have developed a modified one-step procedure for converting terminal alkynes into terminal allenes in higher yields by applying CuI, paraformaldehyde, and dicyclohexylamine.[5f,g] We have also developed a ZnI_2-mediated protocol for the synthesis of 1,3-disubstituted allenes, including chiral examples.[5h–k]

As a type of synthetically attractive functionalized allene, allenols are important synthetic moieties in organic synthesis due to the combination of reactive allene moiety and the hydroxy functionality. Syntheses of 2,5-dihydrofurans,[6] 5,6-dihydro-2*H*-pyrans,[7] epoxides,[8] 3-halo-3-alkenals,[9] and other derivatives[10] from allenols have been reported. Thus, an easy and efficient procedure for obtaining allenols is of high value. Usually, 2,3-allenols can be prepared by the metal-mediated reaction of propargylic bromide with aldehydes[11] or ketones,[12] and the Crabbé reaction of terminal propargylic alcohols.[5a–c] As noted above, compared with the traditional conditions of CuBr and diisopropylamine, allenes can be prepared in higher yields from alkynes and paraformaldehyde with CuI and dicyclohexylamine.[5e,f] Many functional groups, such as mesylate, hydroxy, ether, and amide, are tolerated under these conditions. In this paper, we wish to report our exploration of a general synthesis of basically all allenols, including optically active examples, with different carbon chains between the allene moiety and the hydroxy group (Scheme 1).[5c,13]

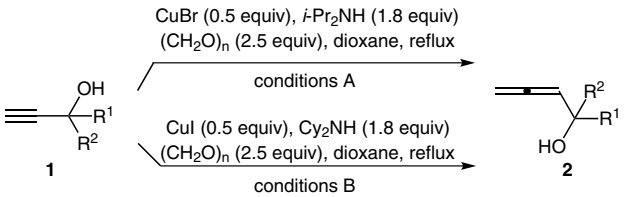
Scope and Limitations

Firstly, we investigated the synthesis of 2,3-allenols using our modified method (Conditions B) and, at the same time, we conducted reactions using the traditional Crabbé reaction conditions as a comparison (Conditions A); the results are summarized in Table 1. For secondary propargylic alcohols with an alkyl substituent at the α-position, the reaction afforded the corresponding allenols **2a** and **2b** in 68% and 77% isolated yield, respectively (Table 1, entries 1 and 2). For propargylic alcohols bearing an aryl group at the α-position, allenols **2c**–**e** were obtained in 75–88% isolated yield (Table 1, entries 3–5). The reaction of tertiary propargylic alcohols substituted with an alkyl and an aryl group at the α-position afforded allenols **2f** and **2g** in 84% and 85% isolated yield, respectively (Table 1, entries 6 and 7). Cyclic alcohols also worked smoothly (Table 1, entries 8–10).

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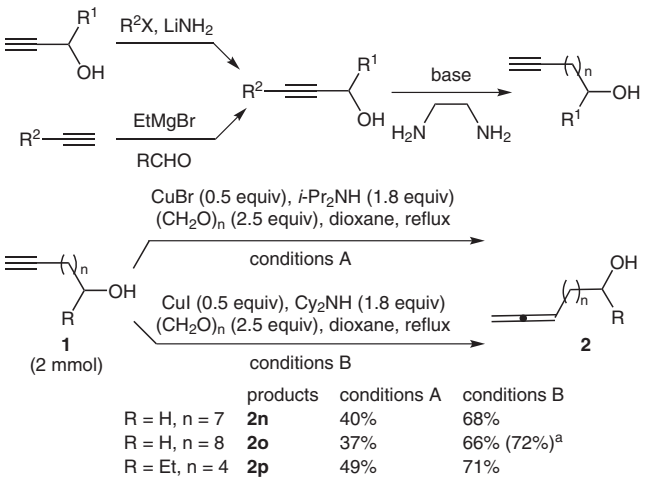
Table 1 Copper(I)-Mediated Synthesis of 2,3-Allenols from Propargylic Alcohols and Paraformaldehyde[a]



Entry	**1**	Conditions A		Conditions B	
	R^1; R^2	Time (h)	Yield (%) of **2**	Time (h)	Yield (%) of **2**
1	H; *n*-C_7H_{15} (**1a**)	2.3	51	2.3	68 (**2a**)
2	H; *n*-C_9H_{19} (**1b**)	2.3	65	2.3	77 (**2b**)
3	H; Ph (**1c**)	2.2	64	3.0	75 (**2c**)
4	H; 4-ClC_6H_4 (**1d**)	2.6	73	2.6	88 (**2d**)
5	H; 4-$MeOC_6H_4$ (**1e**)	2.6	65	2.6	87 (**2e**)
6	Me; 4-Tol (**1f**)	3.3	70	2.3	84 (**2f**)
7	Me; 3-Tol (**1g**)	2.2	65	2.8	85 (**2g**)
8	-$(CH_2)_6$- (**1h**)	3.5	54	2.5	69 (**2h**)
9	-$(CH_2)_5$- (**1i**)	2.3	53	2.5	69 (**2i**)
10	-$(CH_2)_3$- (**1j**)	3.0	51	3.0	65 (**2j**)

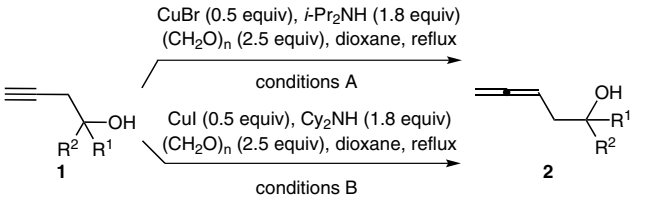
[a] Reactions were carried out on a 2.0-mmol scale in 3.0 mL of 1,4-dioxane.

Secondly, we expanded the scope of substrates to secondary and tertiary homopropargylic alcohols. When the secondary homopropargylic alcohol bears a 4-bromophenyl substituent at the α-position, allenol **2k** was obtained in 64% isolated yield, while homopropargylic alcohol **1l** bearing an electron-rich 4-methoxyphenyl substituent afforded allenol **2l** in 77% isolated yield (Table 2, entries 1 and 2). Tertiary homopropargylic alcohol **1m** was also converted into the corresponding allenol **2m** in 64% isolated yield (Table 2, entry 3).



Scheme 2 Synthesis of allenols **2n**–**p** with different carbon chains between the hydroxy group and the allene moiety. [a] Yield of a 29.6-mmol-scale reaction.

Table 2 Copper(I)-Mediated Synthesis of 3,4-Allenols from Homopropargylic Alcohols and Paraformaldehyde[a]



Entry	**1**	Conditions A		Conditions B	
	R^1; R^2	Time (h)	Yield (%) of **2**	Time (h)	Yield (%) of **2**
1	H; 4-BrC_6H_4 (**1k**)	2.0	39	2.5	64 (**2k**)
2	H; 4-$MeOC_6H_4$ (**1l**)	2.1	37	2.5	77 (**2l**)
3	Me; 4-Tol (**1m**)	3.0	49	2.8	64 (**2m**)

[a] Reactions were carried out on a 2.0-mmol scale in 3.0 mL of 1,4-dioxane.

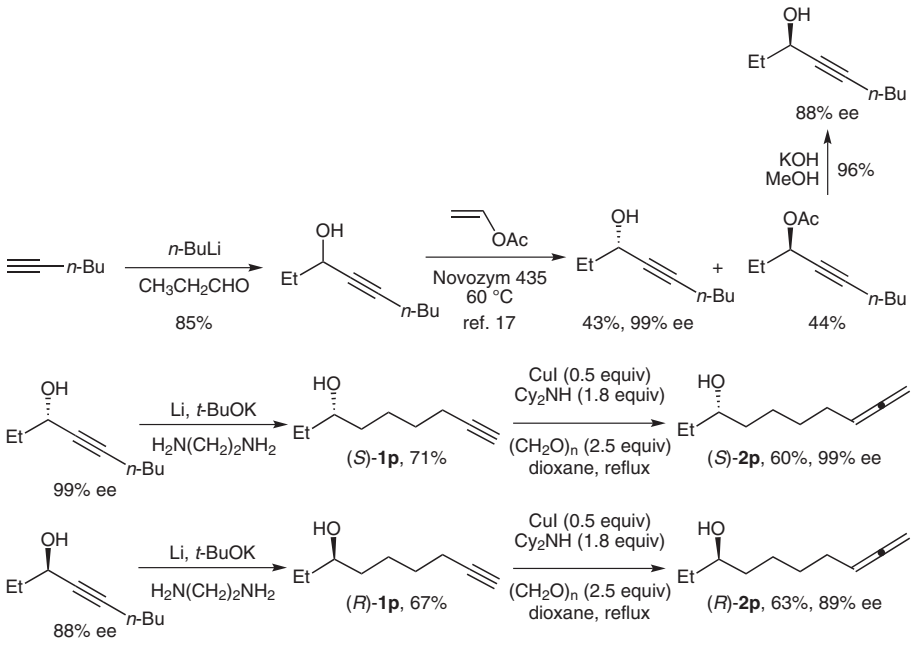
Since ω-alkynols with a long carbon chain can be prepared conveniently from propargylic alcohols by alkylation[14] or from 1-alkynes by deprotonation and 1,2-addition with aldehydes or ketones,[15] and subsequent base-catalyzed triple-bond migration,[16] a series of ω-allenols **2n**–**p** with different carbon chains between the allene moiety and the hydroxy group could be efficiently synthesized using the current protocol (Scheme 2).

Finally, it was observed that optically active ω-allenols such as (*R*)-**2p** and (*S*)-**2p** could be prepared without racemization[17] (Scheme 3).

It can be seen that the replacement of diisopropylamine by dicyclohexylamine and of CuBr by CuI improved the reaction yields, and almost all types of terminal allenols have been synthesized. Of course, it should be noted that this procedure only works for terminal allenols.[13h–j]

In conclusion, we have demonstrated a very general method for the preparation of essentially any terminal allenols from the corresponding alkynols, which may be easily available from propargylic alcohols by alkylation, or from terminal alkynes by deprotonation and 1,2-addition with aldehydes or ketones, and subsequent base-catalyzed triple-bond migration, by applying 0.5 equivalents of CuI, 1.8 equivalents of dicyclohexylamine, and 2.5 equivalents of paraformaldehyde. Due to the diverse utility of allenols in organic chemistry and the broad generality, this method will be of interest since it is easy to undertake and of high efficiency. Further studies in this area are being pursued in our laboratory.

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Scheme 3 Copper(I)-mediated synthesis of optically active allenols from optically active alkynols and paraformaldehyde

^{1}H and ^{13}C NMR spectra were recorded in $CDCl_3$ with a Bruker instrument operated at 300 MHz for ^{1}H NMR and 75 MHz for ^{13}C NMR. Chemical shifts (δ) are given in parts per million (ppm). Infrared spectra were recorded on a Bruker FT-IR spectrophotometer. Mass spectra were recorded on an Agilent spectrometer in EI mode. HRMS spectra were recorded on a Waters spectrometer in EI mode. Cy_2NH, *i*-Pr_2NH, CuI, CuBr, and $(CH_2O)_n$ were purchased from Sinopharm Chemical Reagent Co., Ltd. and were used without further treatment. 1,4-Dioxane was dried over sodium wire with benzophenone as indicator and freshly distilled before use.

Dodeca-10,11-dien-1-ol (2o); Typical Procedure for Conditions A

$(CH_2O)_n$ (149.9 mg, 5.0 mmol), CuBr (143.8 mg, 1.0 mmol), alkynol **1o** (336.6 mg, 2.0 mmol), *i*-Pr_2NH (363.6 mg, 3.6 mmol), and 1,4-dioxane (3 mL) were added sequentially into a dried reaction tube equipped with a reflux condenser under a nitrogen atmosphere. The resulting mixture was stirred under reflux. When the reaction was complete (after 2.4 h, as monitored by TLC), the mixture was cooled to r.t. and filtered. Et_2O (10 mL) and H_2O (5 mL) were added. Then, several drops of concd aq HCl were added to separate the organic layer from the aqueous layer clearly. The aqueous solution was separated and then extracted with Et_2O (3 × 10 mL). The combined organic layer was then dried (anhyd $MgSO_4$). Filtration, concentration, and column chromatography on silica gel (petroleum ether–EtOAc, 10:1) afforded allenol **2o** as a liquid; yield: 134.2 mg (37%).

Dodeca-10,11-dien-1-ol (2o); Typical Procedure for Conditions B

$(CH_2O)_n$ (150.1 mg, 5.0 mmol), CuI (190.8 mg, 1.0 mmol), alkynol **1o** (336.6 mg, 2.0 mmol), Cy_2NH (651.8 mg, 3.6 mmol), and 1,4-dioxane (3 mL) were added sequentially into a dried reaction tube equipped with a reflux condenser under a nitrogen atmosphere. The resulting mixture was stirred under reflux. When the reaction was complete (after 2.4 h, as monitored by TLC), the mixture was cooled to r.t. and filtered. Et_2O (10 mL) and H_2O (5 mL) were added, and then the aqueous solution was separated and extracted with Et_2O (3 × 10 mL). The combined organic layer was then dried (anhyd $MgSO_4$). Filtration, concentration, and column chromatography on silica gel (petroleum ether–EtOAc, 10:1) afforded allenol **2o**[5e] as a liquid; yield: 243.2 mg (66%).

IR (neat): 3330, 2925, 2854, 1956, 1463, 1057 cm^{-1}.

^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.08 (quintet, *J* = 6.8 Hz, 1 H, C=CH), 4.64 (dt, *J* = 6.6, 3.3 Hz, 2 H, C=CH_2), 3.69–3.57 (m, 2 H, OCH_2), 2.05–1.91 (m, 2 H, C=CCH_2), 1.63–1.48 (m, 2 H, CH_2), 1.47–1.21 (m, 13 H, OH and 6 × CH_2).

^{13}C NMR (75.4 MHz, $CDCl_3$): δ = 208.4, 90.0, 74.5, 63.0, 32.8, 29.5, 29.4, 29.3, 29.1, 29.0, 28.2, 25.7.

MS (EI, 70 eV): *m*/*z* (%) = 165 (19.37) [M^+ – OH], 41 (100).

Compounds **2a**–**p** were prepared according to the typical procedures for conditions A and B, and the experimental data are given in the supporting information.

Dodeca-10,11-dien-1-ol (2o); Procedure for Large-Scale Reaction

To a 250-mL three-neck flask equipped with a reflux condenser were added $(CH_2O)_n$ (2.2253 g, 74.0 mmol), CuI (2.8193 g, 14.8 mmol), alkynol **1o** (4.9951 g, 29.6 mmol), 1,4-dioxane (20 mL, to rinse the alcohol), Cy_2NH (9.6467 g, 53.5 mmol), and 1,4-dioxane (22 mL, to rinse the amine). The resulting mixture was refluxed with stirring. When the reaction was complete (after 3 h, as monitored by TLC), the resulting mixture was cooled to r.t. and filtered. The filtrate was acidified with concd HCl to pH 1–2. Then, Et_2O (20 mL) was added to allow the separation of the organic layer from the aqueous layer. The aqueous layer was extracted with Et_2O (3 × 10 mL). The combined organic layer was washed with brine until pH 6–7 and dried (anhyd $MgSO_4$). Filtration, concentration, and column chromatography on silica gel (petroleum ether–EtOAc, 10:1) afforded the terminal allenol **2o**; yield: 3.8875 g (72%).

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